

Experiment 2

Objective: Experimental verification of following electrical circuit problems using microcap.

- (i) Kirchhoff's current law
- (ii) Kirchhoff's voltage law

Apparatus required: Battery, multimeter, Resistors and potentiometers.

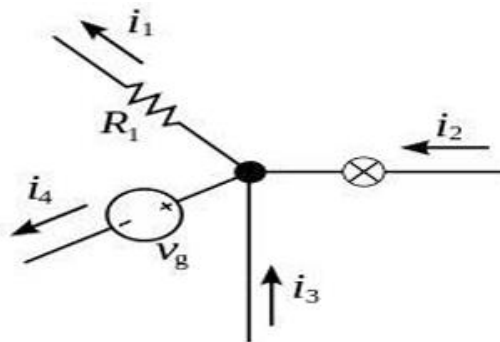
Theory:

Kirchhoff's Current Law

- At any node (junction) in an electrical circuit, the sum of currents flowing into that node is equal to the sum of currents flowing out of that node.
- The algebraic sum of currents in a network of conductors meeting at a point is zero.

$$\sum_{k=1}^n I_k = 0$$

N is total no. of branches with currents towards and away from the node.

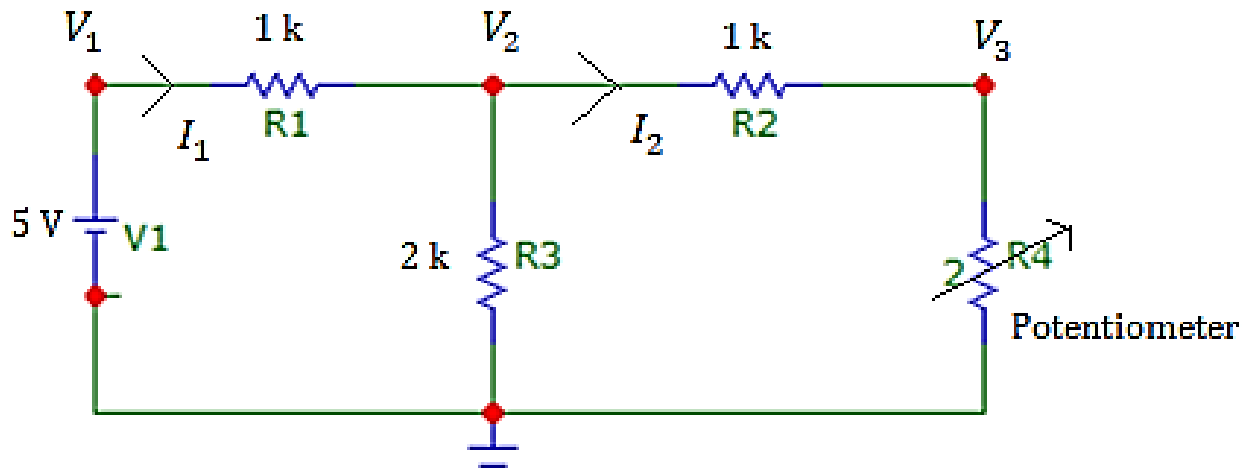


Kirchhoff's voltage Law

Any closed loop in a circuit, the algebraic sum of the potential differences across all elements is zero.

$$\sum_{k=1}^n V_k = 0$$

Circuit Diagram:



PROCEDURE:

1. Connection made as per the circuit diagram.
2. Switch the ON the supply
3. The potential divider is gradually varied in steps and each step the reading of ammeters noted down.
4. The readings are tabulated.

TABULATION (HARDWIRED)

Verification of Kirchoff's Current Law

S.N.	Resistance of Potentiometer (ohm)	V_1	V_2	V_3

Verification of Kirchoff's Voltage Law

S.N.	Resistance of Potentiometer (ohm)	I_1	I_2

RESULT: Hence the Kirchoff's laws were verified using both hardwired components and Microcap.